Introduction

Contributors

This section provides an overview of the student members of the team; their roles; and the clients and supervisors.

Team Members

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Background:
Turkey
BSc Electrical Engineering, TU Eindhoven, The Netherlands

Interests:

Artemi Kurski
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Background:
Estonia
BSc Architecture, University of Bath, United Kingdom

Interests: computational modelling of terrains and the built environment

Ruben Vons
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Background:
The Netherlands
BSc Aeronautical Engineering, Inholland, The Netherlands

Interests: Solving programming challenges and creating nice visuals.

Roles

Client & Supervision Team

The clients of this project are Rijkswaterstaat and Deltares, represented by Daan van der Heide and Maarten Pronk respectively. Additionally, Gina Stavropoulou is representing 3D Geoinformation Group of TUDelft and serves as the liaison between the clients and the university. The supervisors

TODO: Check

provide feedback on the team's work, help define scope and desired outcomes, and share their experience with existing approaches to the problem.

Rijkswaterstaat are a government agency for infrastructure and water management of the Netherlands. Originally created in 1798 for flood prevention, it now also oversees the construction and maintenance of national infrastructure. Some of the notable projects completed by the agency include the Afsluitdijk, constructed between 1920 and 1932, and the Delta Works, constructed between 1954 and 1997. Flooding still remains a key concern for Rijkswaterstaat.

Deltares are an independent research institute specialising in water and subsurface research. Their key mission is 'Enabling Delta Life', which encompasses five key areas focusing on health, safety, and sustainability of life in river deltas. Creating a more accurate river model contributes to several of these areas — in particular, "Safer from flooding" and "Healthy water systems".

Project Definition

Research and Subquestions

The main research question of this project is:

How to create a harmonised cross-border DEM of the Rhine catchment area in an automated (and efficient) way?

As described in the introduction, the project focusses on the issues presented by the cross-border data of the Rhine catchment area. This translates to the creation of an automated pipeline and the report describing the issues that come with attempting the creation of a harmonised DEM. The efficiency refers to creating a pipeline that accounts for the large data quantity and limited computational resources. These two aspects of the main question can be broken down into the following subquestions:

- What is the internationally recognised border of the Rhine catchment area?
- What are the issues with cross-border data?
- What are the INSPIRE requirements for DEMs?
- What are the existing DEMs (local/global)?
- What, if any, are the existing workflows for processing point clouds from heterogeneous sources?
- What are the unknowns/variables in the data we need to consider?
- What are the limitations of an automated harmonisation pipeline?
- How can we test the accuracy of the harmonised DEM?

Answering these subquestions will help to answer the main research question. The first five subquestions will be evaluated in the research phase of the project and relate mostly to existing works and available data. Answering these questions is important as input into the software development and data collection phase. The next research questios will be answered during the software development phase. This question regarding the unknowns will be solved organically by deciding which variables are and are not relevant for the pipeline. The final two questions relate to testing and limitations of the pipeline. These will be answered by verification of the requirements and validation with the client based on findings during testing.

Relevant courses from the MSc Geomatics program

This project applies knowledge gained from several courses from the MSc Geomatics program. Having a basic overview of these courses will highlight the current knowledge possessed by the team. These are the following courses:

- GEO1000 - Python for Geomatics;
- GEO1001 - Sensing Technologies;
- GEO1002 - GIS and Cartography;
- GEO1004 - 3D modelling for the build environment;
- GEO1015 - Digital Terrain Modelling.

GEO1000 is relevant for programming the pipeline in either Python or C++. GEO1001 was the basis for understanding point cloud data collection and processing. GEO1002 is relevant for understanding the data and how to visualise it. GEO1004 is possibly relevant for working with 3D data and point cloud processing. GEO1015 is applicable as it forms the basis for 2D and 2.5D terrain modelling, shortcomings and processing of DEMs.

Requirements

This chapter describes the requirements for this project. The requirements are divided into data requirements, country requirements, map requirements, technical content requirements, and report requirements. The requirements each have a priority assigned based on the MoSCoW method. This method divides requirements into four categories: Must have, Should have, Could have, and Will not have. Must have requirements are those which are mandatory. Should have requirements are more akin to nice-to-have features Could have requirements are those which have been discussed but are not mandatory. Will not have requirements are more related to project scope and refer to features which will not be part of the project. The requirements are listed in Table 1.

Req ID	Description	MoSCoW
DT-01	Global EU DEM (as ground truth)	Must
DT-02	Rhine Watershed Mask	Must
DT-03	Rhine Bathymetry	Could
CT-01	Netherlands is included	Must
CT-02	Germany is included	Must
CT-03	Belgium is included	Should
CT-04	Switzerland is included	Must
CT-05	France is included	Must
CT-06	Luxembourg is included	Must
CT-07	Austria is included	Must
CT-08	Liechtenstein is included (included in Swiss data)	Must
CT-09	Italy is included	Will not have have
MP-01	DSM map	Must
MP-02	DTM map	Must
MP-03	Land-sea mask	Must
MP-04	Bathymetry mask	Could
MP-05	Nodata mask	Should
MP-06	Point cloud density map	Should
TC-01	Workflow is fully automated	Must
TC-02	Workflow is efficient	Should
TC-03	Workflow is reproducible	Must
RP-01	Report on findings	Must
RP-02	Report details the workflow	Must
RP-03	Report details the issues with cross-border data	Must
RP-04	Report details the limitations of the workflow	Must

Table 1: MoSCoW prioritization of data requirements for the project

Methodology

This chapter will go over the approach and methodology of the research that will be performed for this project. In addition, the quality assurance will be discussed.

Research Approach

This subchapter will discuss the approach used to answer each subquestion. They will each be discussed individually and summarized in {Insert table here}

Research Methodology

Quality Assurance

Planning

Phases

Organisation phase

The organisation phase is the first phase of the project. During this phase, the project will be defined and planned. This phase will be used to define the project scope, objectives, deliverables, and timeline. The project team will be formed and roles and responsibilities will be assigned. The project plan will be created and approved by the stakeholders. This phase will last approximately 2 weeks.

The phase consists of the following tasks:
- Project Initiation Document (PID) creation and approval;
- Project planning and scheduling;
- Meeting with stakeholders to discuss project scope, objectives, deliverables, and timeline.

This report is the PID and includes most information aquired during this first phase. The project plan and schedule can be found at the end of this chapter in the form of a Gantt chart. The meeting minutes can be found on the project repository. While elements discussed in this first phase are not final, they will be used as a basis for the project plan and schedule.

Research phase

The research phase is the second phase of the project. During this phase, existing approaches and methods will be evaluated. Additionally, during this phase, data will be collected and evaluated for fitness of use and quality. It is expected that this phase will last approximately 2 weeks.

The phase consists of the following tasks:
- Literature review of existing approaches and methods;
- Review of CRS transformations and their accuracy;
- Refinement of the methodology and approach based on the literature review;
- Data collection and evaluation for fitness of use and quality.

It is likely that other projects have developed different approaches to the problem of creating a cross-border DEM. This will be the main focus of the literature study. Additionally, since each country has its data in a different CRS, it is important to learn how to transform the data into a common CRS. This will be done by reviewing CRS transformations and their accuracy. The methodology and approach described in this PID is based on an intial assumption of the problem and will be refined based on our findings. Lastly, data for both the ground truth and the point clouds will be collected and evaluated in preparation for the next phase.

Software development and data collection phase

The software development and data collection phase is the third phase of the project. During this phase, the software will be developed and tested. This includes both a pipeline prototype and a production version. It is expected that this phase will last approximately 4 weeks.

The phase consists of the following tasks:
- Development of a pipeline prototype;
- Testing of the pipeline prototype on a data subset;
- Development of a production version of the pipeline;
- Testing of the production version of the pipeline a data subset;
- Testing of the production version of the pipeline on the full dataset;
- Software documentation and user manual creation.

The pipeline prototype will be developed as a proof of concept and will be test on a small data subset. For the production version, the pipeline will be developed to handle the full dataset. The production version will be tested on a small data subset and then on the full dataset. The software documentation and user manual will be created to ensure that the software can be used by others.

Midterm presentation phase

The midterm presentation phase is the fourth phase, but will be executed in parallel with the software development and data collection phase. During this phase, the project will be presented to the stakeholders to show the progress made and to receive feedback. In addition, this opertunity may be used to adjust the project scope or objectives. There is no set duration for this phase.

The phase consists of the following tasks:
- Midterm presentation preparation;
- Midterm report creation.

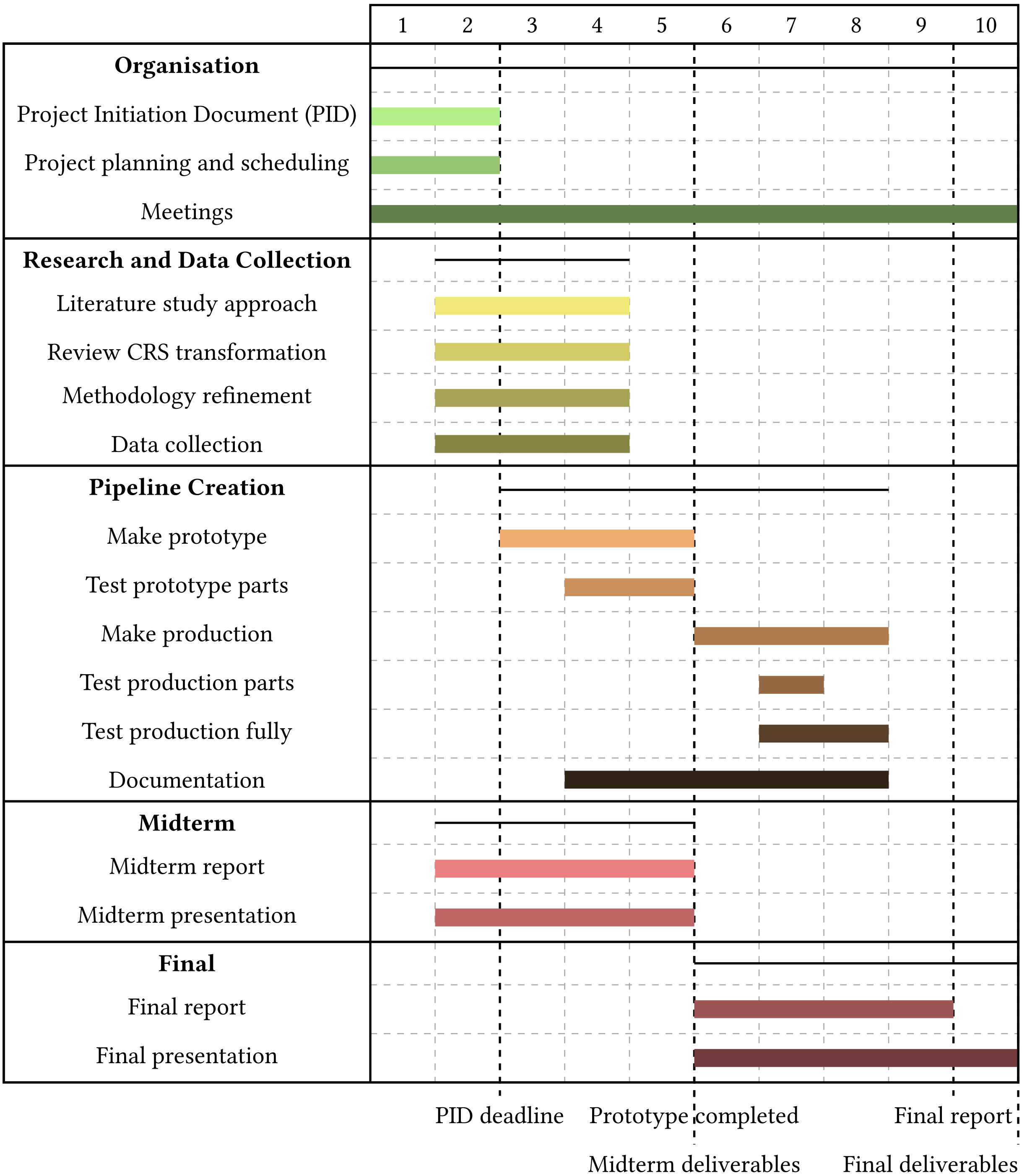
The midterm presentation will be prepared to show the progress made up until that point. The midterm report will be the progress of the final report and will contain the results of the research phase and the software development in its current state. Both the presentation and report will be presented to the stakeholders to receive feedback and to adjust the project scope and/or objectives if neccesary.

Final presentation phase

The final presentation phase is the fifth and final phase of the project. During this phase, the project will be presented to the stakeholders to show the final results and the report. In addition, this will contain the final geomatics day presentation. There is no set duration for this phase.

The phase consists of the following tasks:
- Final presentation preparation;
- Final report creation.

The final presentation will be prepared to show the final results and conclusions of the project. The final report will contain the results of the research phase and the software development in its final state.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Organisation
Project Initiation Document (PID)
Project planning and scheduling
Meetings
Research and Data Collection
Literature study approach
Review CRS transformation
Methodology refinement
Data collection
Pipeline Creation
Make prototype
Test prototype parts
Make production
Test production parts
Test production fully
Documentation
Midterm
Midterm report
Midterm presentation
Final
Final report
Final presentation
PID deadline
Prototype completed
Final report
Midterm deliverables
Final deliverables

Communication

Communication is key to the success of this project. To ensure communication is maintained, both within the team and with stakeholders, regular meetings will be held. The team will meet on a weekly basis to discuss progress, issues and questions. These meetings will be held on Fridays. There will be an additional opertunity to meet on Mondays if neccesary. For each meeting, key talking points will be shared in advance. Meeting minutes will be kept and distributed after each meeting. These meetings will be held online via Microsoft Teams. The team will also communicate internally outside of these meetings, either through Whatsapp or in person.

Risk Analysis

In order to identify and mitigate risks, a risk analysis will be conducted. The method used is the risk matrix method. It is important to mention that this method is not infalible and that it is possible that risks are not identified or that the impact and likelihood are wrongly assessed. However, this method does allow for a more structured approach to risk analysis. Table 2 shows the identified risks, their impact and likelihood, and what can be done to mitigate them.

Likelihood	Harm severity			
	Minor	Marginal	Critical	Catastrophic
Certain	High	High	Very high	Very high
Likely	Medium	High	High	Very high
Possible	Low	Medium	High	Very high
Unlikely	Low	Medium	Medium	High
Rare	Low	Low	Medium	Medium
Eliminated	Eliminated			

Figure 1: Risk Assessment Matrix.

Risk ID	Description	Impact	Likelyhood	Mitigation
1.	Pipeline too computationally taxing or insufficient computational resources.	Critical	Possible	Test with small dataset and adjust spatial extent if necessary.
2.	Insufficient quality/ availability of pointcloud data.	Marginal	Rare	Shift focus to regions with available high-quality data.
3.	CRS transformations are inaccurate.	Critical	Unlikely	Test CRS alignment on select border regions
4.	Missing/conflicting data on border areas.	Critical	Likely	Check spatial overlap of datasets and prioritize one dataset.
5.	Task is too ambitious given timeframe/ team size.	Catastrophic	Unlikely	Check if internal deadlines are met and adjust scope if necessary.
6.	Edge cases not properly assessed due to data quantity and variability.	Minor	Possible	Unideal but acceptable within scope of project.
7.	Task is insufficiently constrained/defined.	Critical	Unlikely	Regular meetings with team and stakeholders to assess and adjust scope and requirements if necessary.
8.	Lacking/inadequate communication within the team and/ or with stakeholders.	Critical	Possible	Schedule regular meetings and maintain open communication channels.
9.	Discovery of unexpected issues.	Marginal	Possible	Regular meetings with team and stakeholders to assess impact and devise mitigation strategies if necessary.
10.	Temporary or permanent absence of team members.	Critical	Rare	Set internal deadlines early to allow for adjustments in case of absence.
11.	Data loss.	Catastrophic	Possible	Implement regular backup procedures.

Table 2: Risk Assessment Table.

https://en.wikipedia.org/wiki/Risk_matrix